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NPS/001/021 – Technical Specification for Paint Systems for Overhead Line Towers

1. Purpose

The purpose of this document is to detail the specification for protective coatings to be used for lattice steel towers on the overhead line distribution networks of CE Electric UK.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
MNT/001/006	1.0	Specification for Tower Paint

2. Scope

This specification provides guidance on the formulation of acceptable two coat urethane alkyd tower paints and detailed requirements for both wet and applied properties. The paint shall meet the specified parameters. In all other respects it must be suitable for application by hand brush to overhead line tower structures as detailed in MNT/001/004 – Specification for the Tower Painting Contract.

3. Technical Requirements

3.1 General Requirements

All paints supplied to the CE Electric UK for use on overhead line towers must meet the following general requirements.

1. Within the normal constraints of good practice all paints should be formulated with the lowest toxicity materials and where possible with high flash point solvents.
2. All paints must be suitable for application over a wide variety of substrates including weathered galvanising, rusty steel, aged paint or any combination of these.
3. Each coat of a multi-coat paint system must be compatible with all preceding coats and, to avoid any dispute, should be supplied by the same manufacturer.
4. All materials must have a shelf life in excess of 12 months. No hard settlement will be acceptable.
5. All materials used on overhead line towers must provide a slip resistant surface suitable for climbing on. The TRRL pendulum test, with RAPRA Four S rubber as a slider, has been identified as a suitable practical objective test for the assessment of slip resistance of painted surfaces. All coats must have a minimum category of 'Low Potential for Slip' according to the limits set by the

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UK Slip Resistance Group, i.e. a measurement of 35 or above under both dry and wet (distilled water) conditions. Measurements to be carried out in accordance with:

- The Measurement of Floor Slip Resistance. Guidelines Recommended by the UK Slip Resistance Group, Issue 2, June 2000.
 - BS 7976 Part 2, 2002. Pendulum testers – Part 2: Method of Operation.
6. Rheological data in the form of a band of acceptable values of viscosity measured using a Rion VT04 Viscometer (No. 1 Rotor, No. 1 cup) from 5°C to 30°C must be available from the paint manufacturer and shall be provided with materials tendered to CE Electric UK for approval on request.

Batch certificates, as shown in Appendix 1, shall be completed by the contractor and submitted to CE Electric UK.

3.2 Canning and Labelling

3.2.1 Containers

For normal brush application the paint shall be supplied in 5 litre metal cans with securely attached flat metal carrying handles.

3.2.2 Labels

Containers shall bear the following information:-

- Manufacturer
- Formulation reference
- Batch number (marled in two positions)
- Date of manufacture
- Quantity
- “Electricity Company Tower Paint”
- Paint type and colour
 - i) Two coat alkyd MIO primer; buff
 - ii) Two coat alkyd top coat; grey
- “Thinning agents must not be used”
- Relevant Health and Safety data.

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3.3 Two Coat Alkyd System – Primer containing MIO

This primer may be used over galvanised steel in various stages of deterioration. It must also be suitable for use over previously painted surfaces from which loose rust and paint has been removed by chipping, scraping and wire brushing.

3.3.1 Composition Requirements

Specific Gravity	1.30-1.40
Pigment content	45-50%w/w
% Zinc phosphate in total pigmentation	40-45% w/w
% lamellar micaceous iron oxide* in total pigmentation	5% w/w
Volatile content	White spirit
Volume solids	48% minimum
Medium	Urethane alkyd 20-25% w/w

*Note: Micaceous Iron Oxide referred to in this specification must comply with ISO 10601 (BS 91/57463), Type 2, Grade 1 but with a minimum thin flake content of 70%.

3.3.2 Other Requirements

(a)	Colour	08 C 35 - 08 E 51 (buff yellow) Matt finish
(b)	Drying time	Surface dry at 20°C 4 hours maximum Hard dry at 20°C 16 hours maximum
(c)	Wet edge time	20°C 10-15 minutes

3.3.3 Application Properties

- (a) To be capable of application by brush direct from the can over the temperature range 5-30°C.
- (b) The paint must be capable of application to give a dry film thickness of 50-75 microns.
- (c) **Viscosity**

ICICone and Plate (20°C)	3.0-4.0 poise
RION VT04 (No 1 Rotor 20°C)	25-35 poise
RION VT04 (No 1 Rotor 10°C)	50 poise maximum
- (d) **Hiding power** Complete obscuration of a Morest Chart at 50 micron dft.
- (e) **Sag resistance**

When applied on vertical surfaces the paint must be sag resistant at 175 microns wet film thickness.
- (f) **Moisture displacement**

Under a normal brushing action the paint should be capable of displacing moisture present in the form of uniformly dispersed droplets which have not coalesced from the substrate without detrimental effect on the dried film.

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3.4 Two Coat Alkyd Systems – Top Coat

This material is to be suitable for application over suitably primed surfaces.

3.4.1 Composition Requirements

Specific Gravity	1.55-1.70
Pigment content	55-60%w/w
% lamellar micaceous iron oxide* in total pigmentation	75% minimum w/w
Volatile content	White spirit
Volume solids	48-52%
Medium	Urethane alkyd 17-20% w/w

*Note: Micaceous Iron Oxide referred to in this specification must comply with ISO 10601 (BS 91/57463), Type 2, Grade 1 but with a minimum thin flake content of 70%.

3.4.2 Other Requirements

(a)	Colour		Tower grey (BS 4800 00 A 09)
(b)	Drying time	Surface dry at 20°C	4 hours maximum
		Hard dry at 20°C	16 hours maximum
(c)	Wet edge time	20°C	10-15 minutes

3.4.3 Application Properties

- (a) To be capable of application by brush direct from the can over the temperature range 5-30°C.
- (b) The paint must be capable of application to give a dry film thickness of 75-100 microns.
- (c) **Viscosity**
 - ICICone and Plate (20°C) 3.5-4.5 poise
 - RION VT04 (No 1 Rotor 20°C) 35-45 poise
 - RION VT04 (No 1 Rotor 10°C) 60 poise maximum
- (d) **Sag resistance**

When applied on vertical surfaces the paint must be sag resistant at 225 microns wet film thickness.
- (e) **Moisture displacement**

Under a normal brushing action the paint should be capable of displacing moisture present in the form of uniformly dispersed droplets which have not coalesced from the substrate without detrimental effect on the dried film.

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4. References

4.1 External Documentation

Reference	Title
BS7976 Part 2 2002	Pendulum testers. Method of operation
ISO 10601 2007	Micaceous iron oxide pigments for paints. Specifications and test methods
BS4800	Schedule of paint colours for building purposes

4.2 Internal documentation

Reference	Title
MNT/001/004	Specification for Tower Painting Contract

5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
S Salkeld	Standards Engineer	S Salkeld	03/09/07

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Bolden	Purchasing Manager	G Bolden	11/09/07

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Standards Manager	G Hammel	11/09/07

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	21/09/07

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards Manager	C Holdsworth	11/09/07

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	14/09/07

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Appendix 1 – Batch Test Certificate for Tower Paints

MANUFACTURER

PRODUCT NAME AND REFERENCE

BATCH NUMBER

DATE OF MANUFACTURE

SPECIFIC GRAVITY

VISCOSITY: CONE & PLATE 20°C

RION VT 04 (20°C, No 1 ROTOR)

RION VT 04 (10°C, No 1 ROTOR)

DRYING TIME (ICI DRYING TIME RECORDER)

SURFACE DRY (20°C)

HARD DRY (20°C)

SAG RESISTANCE

I certify that this material conforms to CE Electric UK Technical Specification NPS/001/021.

SIGNED:..... **DATE:**.....